

Introduction to Statistical Methods

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Measures of Central Tendency and Variability

Descriptive statistics turn a collection of values into a small set of useful summaries. The goal is not to replace the data with one number. It is to describe where values are centered, how widely they vary, what shape they form, and whether any observations deserve closer attention.

[Detailed](#) | [External Sources](#)

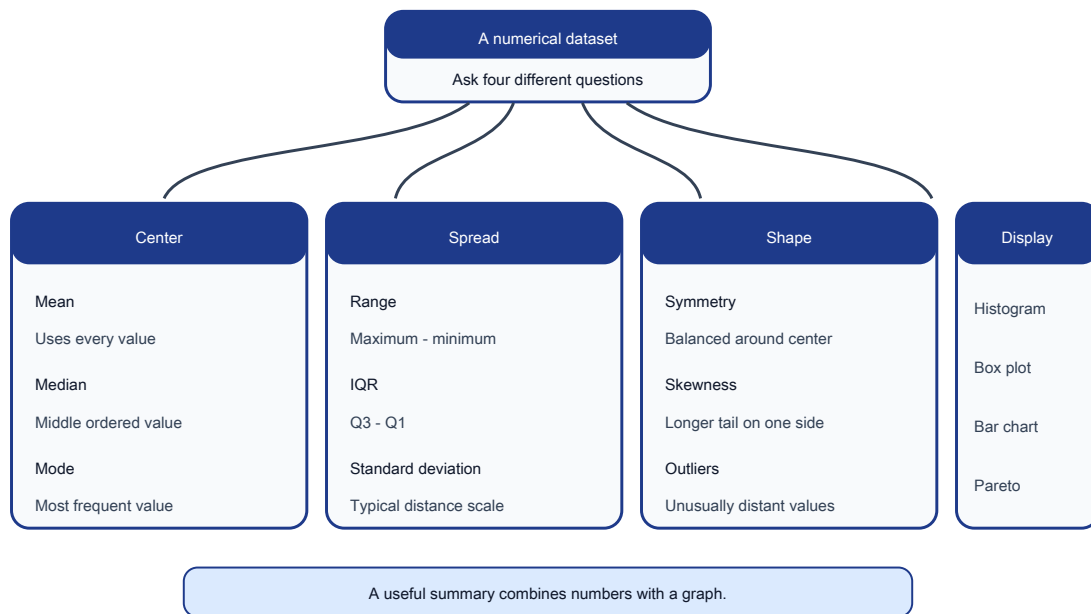


Figure 1. Four complementary views of a numerical dataset. Original course diagram.

Population, Sample, Parameter, and Statistic

A population is the complete group of interest. A sample is the part that is actually observed. A numerical description of a population is a parameter; a numerical description computed from a sample is a statistic.

For example, the mean salary of every employee in a company is a population parameter, conventionally written μ . The mean salary of 50 sampled employees is a sample statistic, conventionally written $\bar{x}$. Statistics are used to learn about parameters, and the resulting inference carries uncertainty.

Measures of Center

Arithmetic Mean

For values $x_1, x_2, \dots, x_n$, the sample mean is:

$$\bar{x} = (1 / n) \sum_{i=1}^n x_i$$

The mean uses every value, so it reacts strongly to extreme observations. A weighted mean is useful when observations have different importance or frequency:

$$\bar{x}_w = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i}$$

Here w_i is the nonnegative weight attached to x_i .

Worked weighted-mean example

Suppose assignments count for 30% of a course grade and an exam counts for 70%. A learner scores 80 on assignments and 60 on the exam. The weighted mean is

$$\bar{x}_w = 0.30(80) + 0.70(60) = 66$$

The ordinary mean $(80 + 60)/2 = 70$ would be inappropriate because the components do not have equal influence.

Median, Mode, and Midrange

- The median is the middle value after sorting. With an even number of observations, it is the mean of the two central values.
- The mode is the most frequent value. A dataset can have no mode, one mode, or several modes.
- The midrange is $(\text{minimum} + \text{maximum}) / 2$. It is easy to calculate but highly sensitive to extremes.

The median is usually more resistant than the mean when a long tail or an outlier is present. That does not make it universally better: the useful measure depends on the question and distribution.

Measures of Variability

Center alone can hide important differences. The sets $\{48, 50, 52\}$ and $\{10, 50, 90\}$ both have mean 50, but the second has much greater spread.

- Range: $\text{maximum} - \text{minimum}$.
- Mean absolute deviation about the mean: $(1/n) \sum |x_i - \bar{x}|$ for a dataset treated descriptively.
- Population variance: [!Population variance averages squared deviations from the population mean.](#)
- Population standard deviation: $\sigma = \sqrt{\sigma^2}$.
- Sample variance: [!Sample variance divides the sum of squared deviations by n minus one.](#)
- Sample standard deviation: $s = \sqrt{s^2}$.

Variance has squared units; standard deviation has the same units as the data. A standard deviation of zero means every value is identical.

Use division by N when the listed observations are the complete population being described. Use division by $n - 1$ for the conventional unbiased estimate of population variance from a sample. The denominator therefore follows the purpose of the calculation, not a cosmetic notation choice.

Five-Number Summary and IQR

The five-number summary contains:

minimum, Q1, median (Q2), Q3, maximum

The interquartile range is:

$$IQR = Q_3 - Q_1$$

It measures the spread of the middle 50% of the ordered data. A common box-plot rule flags potential outliers below $Q_1 - 1.5 IQR$ or above $Q_3 + 1.5 IQR$. The whiskers then extend to the most extreme observed values still inside those fences.

Quartile algorithms differ slightly among textbooks and software, especially for small datasets. State the convention used; the worked example below uses the median-of-halves convention.

Worked Example: Suspicious Transactions

A customer's monthly card transactions, in thousands of rupees, are:

7, 13, 8, 32, 12, 10, 15, 5, 35, 17, 19, 11

- 1 Sort the values: 5, 7, 8, 10, 11, 12, 13, 15, 17, 19, 32, 35.
- 2 The median is $(12 + 13)/2 = 12.5$.
- 3 The lower-half median is $Q_1 = (8 + 10)/2 = 9$.
- 4 The upper-half median is $Q_3 = (17 + 19)/2 = 18$.
- 5 Therefore $IQR = 18 - 9 = 9$.
- 6 The fences are $9 - 1.5(9) = -4.5$ and $18 + 1.5(9) = 31.5$.

The values 32 and 35 exceed the upper fence, so April and September are flagged for investigation. This rule identifies unusual values; it does not prove fraud.

Shape: Symmetry and Skewness

A distribution is symmetric when its two sides have broadly similar shape around the center. It is right-skewed when the longer tail extends toward larger values, and left-skewed when the longer tail extends toward smaller values.

In common unimodal examples, right skew often gives $\text{mean} > \text{median}$, while left skew often gives $\text{mean} < \text{median}$. Treat these as patterns, not definitions. A histogram or box plot should accompany numerical summaries.

The slides introduce Pearson's second coefficient of skewness:

$$S_k = \frac{3(\text{mean} - \text{median})}{\text{standard deviation}}$$

Its sign describes direction under this convention, but one coefficient cannot fully describe an irregular or multimodal distribution.

Distribution-Based Spread Statements

For an approximately normal distribution, the empirical rule places about 68%, 95%, and 99.7% of observations within 1, 2, and 3 standard deviations of the mean.

For any distribution with finite mean and positive finite variance, Chebyshev's theorem guarantees that at least

$$P(|X - \mu| < k\sigma) \geq 1 - \frac{1}{k^2}$$

of values lie strictly within k standard deviations of the mean for $k > 1$. If the variance is zero, every value equals the mean and the distribution is handled directly rather than by standardizing with σ .

Chebyshev's result is a guaranteed lower bound, not an approximation to the exact percentage. At $k = 2$, it guarantees at least 75%; the actual proportion may be larger.

Reading Summaries Together

No single statistic tells the complete story. Consider monthly waiting times of 4, 5, 5, 6, 30 minutes. The median is 5, which describes a typical visit, while the mean is 10, pulled upward by one long delay. Reporting only the median hides the operational failure; reporting only the mean makes most visits seem slower than they were.

A useful description combines:

- a center, such as mean or median;
- a spread, such as standard deviation or IQR;
- a shape or graph showing skewness and clusters;
- the sample size and measurement units;
- an explanation of unusual observations.

This combination lets the reader distinguish ordinary variation from a data-quality problem or a genuinely rare event.

Same center, different predictability

Two delivery services can both average 30 minutes while behaving very differently. One may usually take 28 to 32 minutes; another may alternate between 10 and 50. The mean describes center, but spread describes consistency. Operational decisions therefore need both.

Common Interpretation Mistakes

- Comparing standard deviations for variables measured in different units without standardizing or using a relative measure.
- Assuming the mean must be one of the observed values.
- Treating a box-plot flag as proof that an observation is erroneous.
- Applying the empirical rule to a visibly non-normal distribution.
- Reporting many decimal places that the original measurements cannot justify.

Choosing a Visual Summary

- Use a histogram for the distribution of one numerical variable. Bin choices affect what is visible.
- Use a box plot to compare center, IQR, and potential outliers across groups.
- Use a bar chart for magnitudes across categories. Bars are separated because categories are not numeric intervals.
- Use a Pareto chart when categories should be ordered from most to least frequent, often with cumulative percentage.
- Use a frequency polygon to connect class-midpoint frequencies and an ogive to show cumulative frequency.

A robust first pass is: inspect the raw values, draw a relevant graph, calculate center and spread, identify unusual observations, and interpret every result in the data's units and context.

For derivations, denominator choices, additional examples, diagnostics, and practice, see [Know More](#).

Axioms and Basic Probability Concepts

Probability gives a consistent language for uncertainty. It connects an experiment, its possible outcomes, and the events we care about to numbers between 0 and 1. Those numbers become useful only when the experiment and sample space are defined clearly.

[Detailed](#) | [External Sources](#)

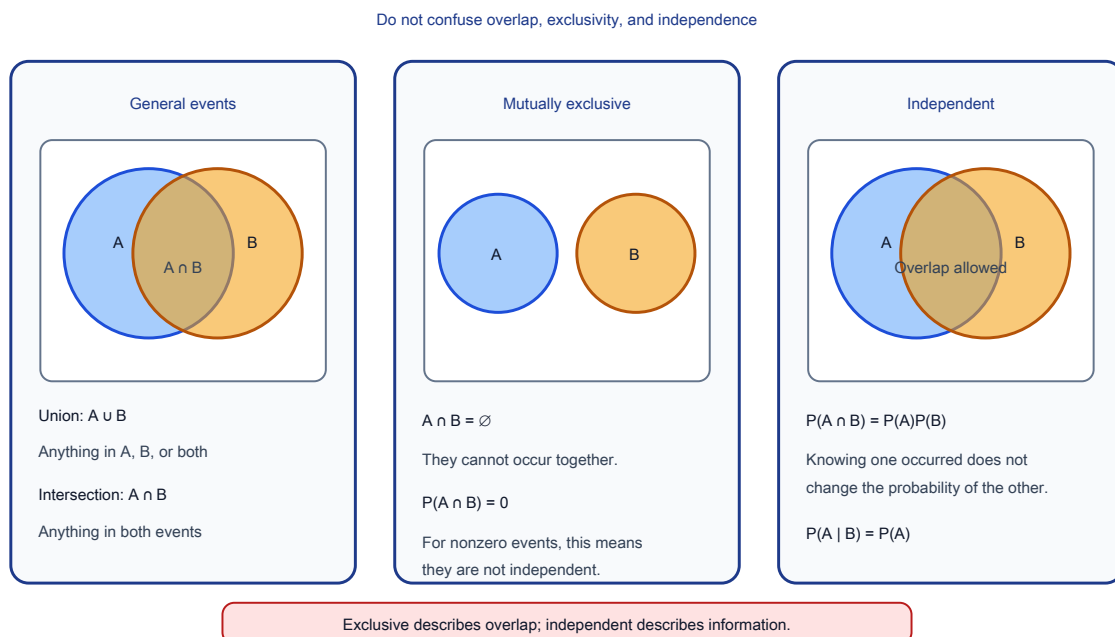


Figure 1. Exclusivity concerns overlap; independence concerns whether information changes probability. Original course diagram.

Experiment, Outcome, Sample Space, and Event

- A random experiment is a repeatable process whose individual outcome is uncertain.
- An outcome is one possible result.
- The sample space S is the set of all possible outcomes.
- An event is a subset of S .

- An elementary event contains exactly one outcome in the chosen sample space.

For one fair die roll, $S = \{1, 2, 3, 4, 5, 6\}$. The event “an even result” is $\{2, 4, 6\}$; it is not elementary because it contains three outcomes. $\{2\}$ is an elementary event.

Under the standard discrete definition, an elementary event contains one outcome. Therefore $\{2\}$ is elementary, while the even-result event $\{2, 4, 6\}$ is a composite event.

Ways to Assign Probability

The slides introduce three practical viewpoints:

- Classical: When a finite sample space has equally likely outcomes, $P(A) = |A| / |S|$.
- Empirical: Relative frequency from repeated observations estimates probability: $\text{number of observed } A \text{ outcomes} / \text{number of trials}$.
- Subjective: A probability can encode informed judgment when repeated data or symmetry are unavailable. It should still obey the probability axioms and should be updated when evidence changes.

The classical ratio is invalid if the outcomes are not equally likely. Historical frequency is an estimate from available data, not a guarantee about the next trial.

The Probability Axioms

For a sample space S and events in that space:

- 1 Non-negativity: $P(A) \geq 0$.
- 2 Normalization: $P(S) = 1$.
- 3 Countable additivity: For pairwise mutually exclusive events $A_1, A_2, \dots$, $P(\cup A_i) = \sum P(A_i)$.

Useful consequences include:

- $P(\emptyset) = 0$.
- $P(A^c) = 1 - P(A)$.
- $0 \leq P(A) \leq 1$.
- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

The subtraction in the union formula prevents the overlap from being counted twice.

Event Relationships

The union $A \cup B$ contains outcomes in A , in B , or in both. The intersection $A \cap B$ contains outcomes common to both. The complement A^c contains outcomes in S but not in A .

Events are mutually exclusive when $A \cap B = \emptyset$, so they cannot occur together. Events are independent when:

$$P(A \cap B) = P(A)P(B)$$

For positive-probability events, this is equivalent to $P(A | B) = P(A)$. Mutually exclusive nonzero events are dependent: learning that one occurred makes the other impossible.

A quick contrast

For one die roll, “even” and “odd” are mutually exclusive because they cannot happen together. They are not independent: once an even result is known, the probability of odd becomes zero.

For two fair coin tosses, “first toss is heads” and “second toss is heads” are independent because learning the first result does not change the second-toss probability. They are not mutually exclusive because both occur in HH.

Counting Possibilities

Probability calculations often begin by counting outcomes.

Product Rule

If one stage has m choices and a following stage has n choices, there are mn ordered combinations. A car with 3 engines, 8 colors, and 4 interior packages has $3 \times 8 \times 4 = 96$ configurations.

Permutations and Combinations

For n distinct objects and r selections:

- Ordered selection: n Permutations count ordered selections using n factorial divided by n minus r factorial.
- Unordered selection: n Combinations count unordered selections using n factorial divided by r factorial times n minus r factorial.

With replacement and n choices at each of r ordered draws, there are n^r sequences. Without replacement, available choices decrease after each draw.

Worked Example: Exactly One Defective Part

A bin contains 2 defective and 4 acceptable parts. Three are selected without replacement.

The total number of unordered samples is:

$$6C_3 = 20$$

To obtain exactly one defective part, select 1 of the 2 defective parts and 2 of the 4 acceptable parts:

$$2C_1 \times 4C_2 = 2 \times 6 = 12$$

Therefore:

$$P(\text{exactly one defective}) = \frac{12}{20} = 0.60$$

This method works because each three-part subset is equally likely under random sampling without replacement.

Worked Example: “At Least One” by Complement

Three independent messages each have probability 0.10 of failing to arrive. Directly listing every case with one, two, or three failures is possible but unnecessarily long. It is simpler to subtract the probability of no failures from one:

$$\begin{aligned}P(\text{at least one failure}) &= 1 - P(\text{no failures}) \\&= 1 - (0.90)^3 = 0.271\end{aligned}$$

The complement strategy is especially useful for phrases such as “at least one,” “not all,” and “one or more.” The multiplication $(0.90)^3$ is valid because the example explicitly assumes independent deliveries.

Sanity Checks

- A union probability cannot be smaller than either event it contains.
- An intersection probability cannot exceed either participating event.
- If two positive-probability events are mutually exclusive, they cannot be independent.
- Probabilities over all mutually exclusive, exhaustive outcomes must sum to one.
- A very detailed sample space is not automatically better; choose outcomes that preserve the distinctions needed by the question.

Common Language Translations

Phrase	Probability structure
“A or B”	union $A \cup B$, including overlap unless stated otherwise
“A and B”	intersection $A \cap B$
“not A”	complement A^c
“at least one”	often easiest as $1 - P(\text{none})$
“exactly one”	one occurs and the others do not
“given B”	restrict the reference set to B

Translate words into events before calculating. This prevents many counting and denominator errors.

A Reliable Problem-Solving Routine

- 1 State the experiment and sample space.
- 2 Define events in words and symbols.
- 3 Decide whether order and replacement matter.
- 4 Check whether outcomes are equally likely before using a counting ratio.

- 5 Use union, intersection, complement, or conditional structure explicitly.
- 6 Check that the answer lies in $[0, 1]$ and makes sense in context.

For derivations, counting diagnostics, and additional problems, see [Know More](#).

Conditional and Total Probability

New information changes the set of possibilities that matters. Conditional probability formalizes that change: instead of measuring an event across the entire sample space, it measures the event inside a stated condition.

[Detailed](#) | [External Sources](#)

Definition

For events A and B with $P(B) > 0$, the probability of A given B is:

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

The condition B becomes the new reference set. This is why the denominator is $P(B)$, not $P(S)$.

Rearranging gives the multiplication rule:

$$P(A \cap B) = P(B)P(A | B) = P(A)P(B | A)$$

Order in the notation matters. $P(A | B)$ and $P(B | A)$ usually differ.

Reading a Contingency Table

The slides use an employment table with 100 male and 55 female employees:

Position	Male	Female	Total
Managerial	8	3	11
Professional	31	13	44
Technical	52	17	69
Clerical	9	22	31
Total	100	55	155

If F means female and T means technical, then conditioning on technical employees restricts the denominator to 69:

$$P(F | T) = \frac{17}{69} \approx 0.246$$

$$P(T | F) = \frac{17}{55} \approx 0.309$$

If the question is reversed, conditioning on female employees gives:

The same joint count appears in both calculations, but the denominators differ.

Joint, marginal, and conditional views

Using the same table:

- $17/155$ is the joint probability of female and technical.
- $69/155$ is the marginal probability of technical.
- $17/69$ is the conditional probability of female among technical employees.

All three use the same cell count, but each answers a different question. State the reference group in words before choosing the denominator.

Independence Through Conditional Probability

Events A and B are independent when knowing B does not change the probability of A :

$$P(A | B) = P(A)$$

provided $P(B) > 0$. Substitution into the multiplication rule gives the equivalent test:

$$P(A \cap B) = P(A)P(B)$$

Do not assume independence merely because events appear unrelated. It is a model claim that should come from the data-generating process or be checked against probabilities.

The Law of Total Probability

Suppose $B_1, B_2, \dots, B_k$ form a partition of S : they are mutually exclusive and together cover the sample space. Then for any event A :

$$P(A) = \sum_{i=1}^k P(B_i)P(A | B_i)$$

Each term is one disjoint path to A . Adding the path probabilities gives the marginal probability of A .

Worked Example: Fuel Choice and Full Tanks

At a fuel station:

- 40% choose Regular, and 30% of Regular customers fill the tank.
- 35% choose Plus, and 60% of Plus customers fill the tank.
- 25% choose Supreme, and 50% of Supreme customers fill the tank.

Let F be “fills the tank.” The fuel types form a partition, so:

$$\begin{aligned}
P(F) &= P(R)P(F | R) + P(P)P(F | P) + P(S)P(F | S) \\
&= 0.40(0.30) + 0.35(0.60) + 0.25(0.50) \\
&= 0.120 + 0.210 + 0.125 \\
&= 0.455
\end{aligned}$$

Thus 45.5% of customers are expected to fill the tank under the stated model.

The joint probability of choosing Plus and filling is one branch:

$$P(P \cap F) = 0.35(0.60) = 0.21$$

The difference between 0.21 and 0.60 is important. The first is the probability that a randomly selected customer both chooses Plus and fills the tank. The second is the probability of filling after the customer is already known to have chosen Plus.

Tables and Trees Express the Same Structure

A contingency table is convenient when joint counts or proportions are already available. A probability tree is convenient when an experiment unfolds in stages. In a tree:

- multiply probabilities along a branch to obtain a joint probability;
- add mutually exclusive branches to obtain a marginal probability;
- divide a relevant joint probability by the conditioning probability to obtain a conditional probability.

Common Errors

- Reversing $P(A | B)$ and $P(B | A)$.
- Dividing by the full sample size after a condition has restricted the reference group.
- Multiplying $P(A)P(B)$ without establishing independence.
- Adding branch probabilities that are not mutually exclusive.
- Forgetting the requirement $P(B) > 0$ in $P(A | B)$.

Real Application: Interpreting Screening Results

Suppose 2% of a population has a condition, 90% of affected people test positive, and 5% of unaffected people also test positive. The total positive-test probability is

$$\begin{aligned}
P(+) &= 0.02(0.90) + 0.98(0.05) \\
&= 0.018 + 0.049 = 0.067
\end{aligned}$$

The joint probability of having the condition and testing positive is 0.018. Therefore the fraction of positive tests that come from affected people is:

$$P(\text{condition} | +) = \frac{0.018}{0.067} \approx 0.269$$

Even with a sensitive test, a positive result does not imply a 90% chance of disease. The 90% number conditions on already having the disease; the practical question conditions on seeing a positive result. This reversal leads directly to Bayes theorem in [Know More](#).

Takeaway

Conditioning changes the denominator because it changes the population being considered. Multiply along a path for a joint probability, add disjoint paths for a marginal probability, and never reverse a condition without a Bayes calculation.

For a full table derivation, total-probability diagnostics, and practice, see [Know More](#).

Bayes Theorem and Naive Bayes

Bayes theorem reverses a conditional question. It combines what was believed before an observation with how compatible that observation is under each possible cause. The result is an updated, or posterior, probability.

[Detailed](#) | [External Sources](#)

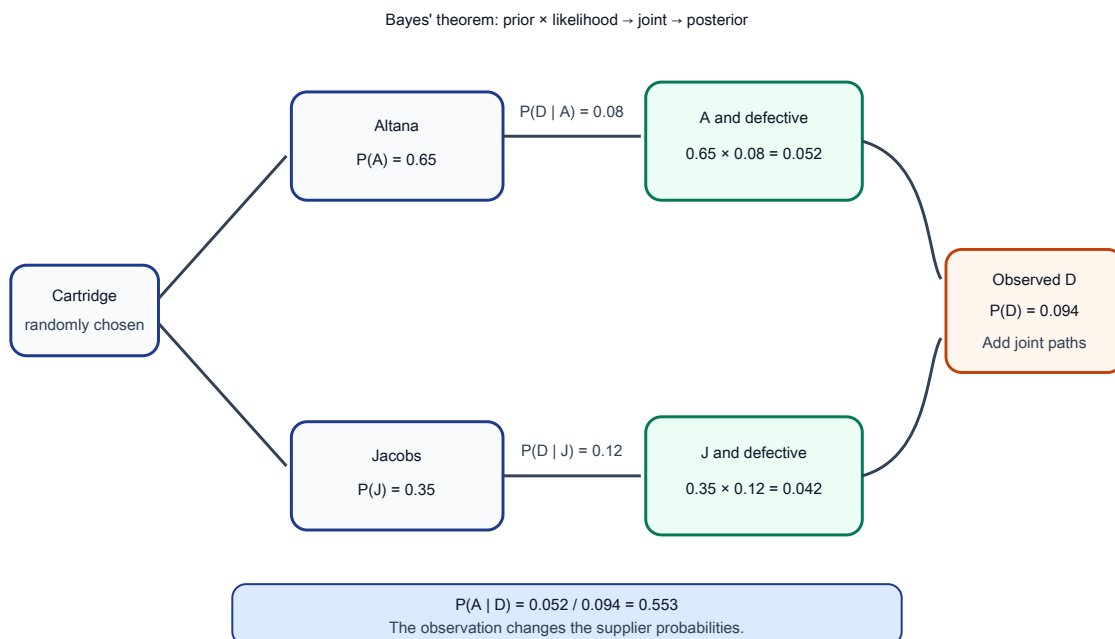


Figure 1. Prior probabilities flow through defect likelihoods to joint and posterior probabilities. Original course diagram.

Bayes Theorem for Two Events

For events A and B with $P(B) > 0$:

$$P(A | B) = \frac{P(B | A)P(A)}{P(B)}$$

The terms have distinct roles:

- $P(A)$ is the prior probability of A .
- $P(B | A)$ is the likelihood of observing B if A holds.
- $P(B)$ is the evidence or marginal probability of B .
- $P(A | B)$ is the posterior probability after observing B .

Bayes theorem does not say that $P(A | B) = P(B | A)$. It shows exactly how the two are related.

Derivation from Conditional Probability

The multiplication rule writes the same joint event in either conditional direction.

$$P(A \cap B) = P(A)P(B | A)$$

Equivalently, $P(A \cap B) = P(B)P(A | B)$. Equating the expressions and dividing by $P(B)$ gives Bayes theorem.

If $B_1, \dots, B_k$ partition the sample space, then:

$$P(B_j | A) = \frac{P(B_j)P(A | B_j)}{\sum_{i=1}^k P(B_i)P(A | B_i)}$$

The denominator is the law of total probability. It adds every mutually exclusive path by which A can occur.

Worked Example: Defective Cartridge

Altana supplies 65% of cartridges and Jacobs supplies 35%. Their defect rates are 8% and 12%, respectively. A purchased cartridge is defective. What is the probability that Altana supplied it?

Let A mean Altana, J mean Jacobs, and D mean defective.

$$P(A \cap D) = 0.65(0.08) = 0.052$$

$$P(J \cap D) = 0.35(0.12) = 0.042$$

$$P(D) = 0.052 + 0.042 = 0.094$$

Therefore:

$$P(A | D) = \frac{0.052}{0.094} \approx 0.553$$

Although Altana supplies 65% of all cartridges, it accounts for only about 55.3% of defective cartridges because Jacobs has the higher defect rate.

Reading the update

Observing a defect moves Altana's probability from the prior 0.65 down to the posterior 0.553. This direction is sensible because a defect is relatively more likely under Jacobs. Bayes theorem combines that likelihood evidence with Altana's larger supply share.

From Bayes Theorem to Classification

Suppose a class variable is C and a record has features $X = (x_1, \dots, x_n)$. Bayes theorem gives:

$$P(C | X) = \frac{P(C)P(X | C)}{P(X)}$$

Naive Bayes assumes the features are conditionally independent given the class:

$$P(X | C) = \prod_{i=1}^n P(x_i | C)$$

For classification:

$$\text{score}(C) = P(C) \prod_{i=1}^n P(x_i | C)$$

Choose the class with the largest score. The common denominator $P(X)$ is not needed for comparing classes because it is the same for every candidate class.

The assumption is conditional independence given the class, not ordinary independence among features across the entire dataset. Two symptoms may be associated overall because both are related to the same disease, yet a naive Bayes model assumes that after the disease class is fixed, one symptom provides no further information about the other.

Worked Naive Bayes Decision

The slides classify a customer with:

Age = Young, Income = Medium, Student = Yes, Credit = Fair

The two prior-times-likelihood scores are:

$$\text{score(Yes)} = \frac{9}{14} \times \frac{2}{9} \times \frac{4}{9} \times \frac{6}{9} \times \frac{6}{9} \approx 0.0282$$

$$\text{score(No)} = \frac{5}{14} \times \frac{3}{5} \times \frac{2}{5} \times \frac{1}{5} \times \frac{2}{5} \approx 0.00686$$

Since $0.0282 > 0.00686$, the classifier predicts **Yes**. These are unnormalized scores. Dividing each by their sum would produce posterior probabilities within this two-class model.

Practical Caveats

- A zero training count can make an entire product zero. Additive smoothing assigns a small nonzero probability to unseen feature values.
- Multiplying many small probabilities can underflow numerically. Implementations usually add log-probabilities instead.
- The largest posterior score may classify well even when the probability estimate is poorly calibrated.
- Correlated features can violate the naive assumption and effectively count similar evidence more than once.

Additive smoothing example

Suppose class **C** contains training documents in which the word “refund” has never appeared. An unsmoothed estimate gives probability zero, forcing the complete class score to zero whenever that word appears. With add-one smoothing over a vocabulary of **V** words,

$$P(\text{word} \mid C) = \frac{\text{count}(\text{word}, C) + 1}{\text{total words in } C + V}$$

The unseen word receives a small probability rather than impossible status. Smoothing expresses uncertainty from limited data; it does not claim that all words are equally common.

Multinomial and categorical naive Bayes commonly use additive smoothing. This avoids zero products but does not guarantee well-calibrated probabilities, so use held-out calibration checks when the numerical posterior itself drives a decision.

Why Log Scores Are Used

For many features, multiplying probabilities such as $0.02 \times 0.04 \times 0.01$ can become smaller than floating-point arithmetic can represent accurately. Taking logarithms converts products into sums:

$$\log \text{score}(c) = \log P(c) + \sum_{i=1}^n \log P(x_i \mid c)$$

The logarithm is strictly increasing, so the class with the largest product also has the largest log score. This changes the computation, not the model's decision rule.

When Naive Bayes Is Useful

Naive Bayes is attractive when training data are limited, features are numerous, and a fast baseline is needed. Text classification is a common example: word-count features are not truly independent, but the simplified model can still rank classes effectively. Its assumptions should be judged by predictive performance and decision needs rather than by whether they are literally true in every detail.

Final Checklist

- 1 Define the competing causes or classes.
- 2 Verify that their priors form an appropriate partition.
- 3 Use likelihoods in the correct direction.
- 4 Compute all joint paths contributing to the evidence.
- 5 Normalize when an actual posterior probability is required.
- 6 Check whether the result is plausible relative to both the prior and the observed evidence.

Takeaway

Bayes theorem updates a prior using likelihood evidence and normalizes across possible causes. Naive Bayes turns this rule into a fast classifier by factorizing the feature likelihood. Smoothing, log-space computation, feature dependence, and calibration all affect practical interpretation.

For the partition proof, log-score implementation, smoothing, diagnostics, and practice, see [Know More](#).

Random Variables and Joint Distributions

A probability model describes uncertain outcomes. A random variable turns each outcome into a number that can be analyzed. For example, instead of recording every detail of a support shift, let X be the number of urgent tickets received in one hour.

[Detailed](#) | [External Sources](#)

From Outcomes to Numbers

A random variable is a function from the sample space to real numbers. The capital letter X names the rule; a lowercase x denotes one possible value.

This distinction matters. Probability belongs to events such as $X = 2$ or $X \leq 2$, not to the symbol X by itself.

Discrete Random Variables and PMFs

A discrete random variable has a finite or countably infinite set of possible values. Counts—customers arriving, failed components, or correct answers—are common examples.

Its probability mass function (PMF) is

$$p(x) = P(X = x)$$

A valid PMF satisfies two checks:

- $p(x) \geq 0$ for every possible value;

- the probabilities over all possible values sum to 1.

Suppose the hourly urgent-ticket count has this PMF:

x	0	1	2	3
P(X=x)	0.20	0.40	0.30	0.10

Then $P(X \geq 2) = 0.30 + 0.10 = 0.40$. The values 0,1,2,3 are outcomes of the random variable; their probabilities are masses attached to those values.

Continuous Random Variables and PDFs

A continuous random variable can take values across an interval, such as waiting time or temperature. Its probability density function (PDF), written $f(x)$, places probability over intervals.

The total area under a valid density is 1, and the density is never negative. For $a < b$, $P(a < X < b)$ is the area under $f(x)$ from a to b .

For a continuous variable, a single exact point has probability zero:

$$P(X = a) = 0$$

This does not mean the value is impossible in ordinary language. It means a point has zero width and therefore zero area under a continuous density. Consequently, strict versus non-strict endpoints make no probability difference for a continuous distribution.

Do not interpret a density height as a point probability. A density may exceed 1; only its area must total 1.

Cumulative Distribution Function

The cumulative distribution function (CDF) works for both discrete and continuous variables:

$$F(x) = P(X \leq x)$$

It is nondecreasing, approaches 0 far to the left, and approaches 1 far to the right. For a discrete variable, it jumps at values with positive mass. For a continuous variable with a smooth density, it accumulates area.

Mean and Variance of a Random Variable

The expected value or mean is the probability-weighted long-run average. For a discrete random variable,

$$E(X) = \mu = \sum x p(x)$$

For the ticket example,

$$E(X) = 0(0.20) + 1(0.40) + 2(0.30) + 3(0.10) = 1.30$$

The interpretation is not “every hour has 1.3 tickets.” Across many comparable hours, the average count tends toward 1.3 under this model.

Variance measures squared distance from that mean:

$$\text{Var}(X) = E[(X - \mu)^2]$$

For this example, $E(X^2)=2.50$, so $\text{Var}(X)=2.50-1.30^2=0.81$. Standard deviation is $\sqrt{0.81}=0.90$ tickets. For continuous variables, corresponding expectations use integrals rather than sums.

Two Random Variables Together

A joint distribution assigns probabilities to pairs such as (X,Y) . For example, X may count urgent tickets and Y may record whether an on-call engineer was available.

In a discrete joint probability table, every cell is $P(X=x, Y=y)$. Row or column sums give marginal probabilities for one variable. Dividing a joint cell by the relevant marginal gives a conditional probability. Thus joint distributions connect the earlier table-based probability ideas to random variables.

Two random variables are independent when every joint probability factors into the product of its marginals:

$$P(X=x, Y=y) = P(X=x)P(Y=y)$$

One matching cell is not enough; the condition must hold across all possible pairs.

Common Mistakes

- Confusing a random variable with one observed value.
- Using a PDF height as though it were $P(X=x)$ for a continuous variable.
- Forgetting that PMF values sum to 1, while PDF probability comes from area.
- Interpreting an expected value as a value that must occur.
- Computing a marginal from only one joint-table cell instead of summing the full row or column.

Takeaway

A random variable translates uncertain outcomes into numbers. Discrete variables use probability masses; continuous variables use density over intervals; the CDF accumulates probability in either case. Expected value describes long-run center, variance describes spread, and a joint distribution records how two random variables behave together.

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